

GlobeTrotter

Empowering Personalized Travel Planning

Project Overview

GlobeTrotter is a sophisticated, centralized travel planning platform designed to transform the chaotic process of multi-city trip organization into a seamless, guided, and editorial-style experience. By unifying destination discovery, itinerary building, activity curation, and budget management, GlobeTrotter serves as the single source of truth for the modern traveler.

THE PROBLEM STATEMENT

Contemporary travel planning is characterized by extreme fragmentation. Information is typically scattered across disparate tools—notes apps for ideas, maps for geography, spreadsheets for budgeting, and calendars for scheduling. This creates a high cognitive load and significant friction during the planning phase.

PROJECT OBJECTIVES

Streamlined Architecture

Provide a guided "Trip Architect" that simplifies complex decision-making.

Centralized Discovery

Enable users to search for cities and tourist attractions within the interface.

Financial Clarity

Integrate target budgeting to ensure itineraries remain financially viable.

Collaboration & Sharing

Facilitate the sharing of professional-grade itineraries via public links.

Solution & Core Features

GlobeTrotter addresses travel complexity through a "Guided Architect" approach. Rather than presenting a blank canvas, the system leads the user through a structured funnel—from high-level aspirations to granular daily activities.

IMPLEMENTED CORE FEATURES

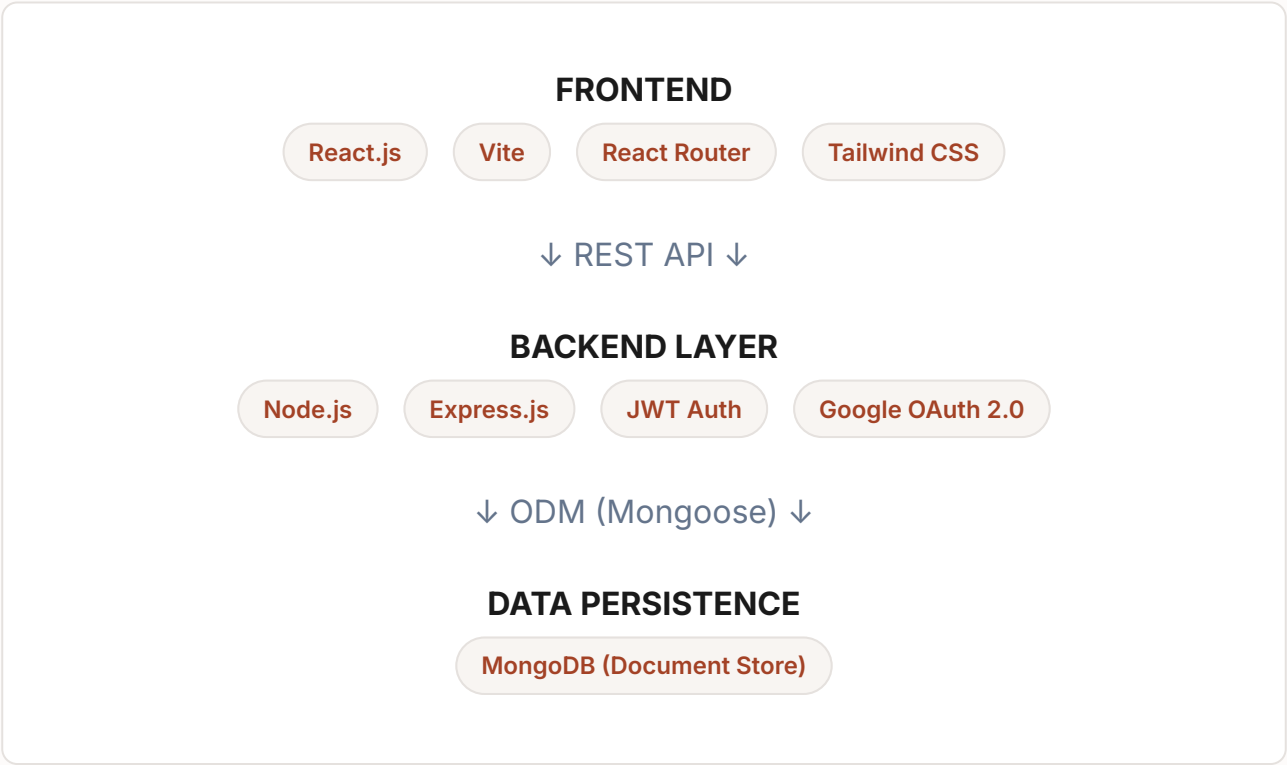
- **Dashboard & My Trips:** A high-level overview of the user's travel history and upcoming journeys.
- **Guided Trip Architect:** A multi-step wizard (Itinerary Builder) that captures trip metadata and dates.
- **City & Destination Selection:** Integrated discovery allowing users to add multiple cities to a single trip.
- **Activity & Tourist Place Curation:** A dedicated interface to attach specific local experiences to trip stops.
- **Budget & Privacy Controls:** Financial target setting and granular visibility toggles (Private vs. Public).
- **Secure Onboarding:** JWT authentication and Google OAuth 2.0 for frictionless access.

USER JOURNEY MAP

- 01. Access:** User authenticates via JWT or Google OAuth.
- 02. Initiation:** User triggers the "Plan a Trip" workflow from the Dashboard.
- 03. Architecture:** The Guided Architect captures Trip Name, Dates, and Narrative.
- 04. Discovery:** User selects cities and filters for specific tourist places.
- 05. Optimization:** Financial targets are set, and the journey summary is reviewed.
- 06. Finalization:** Itinerary is saved and a public shareable link is generated.

System Architecture & Technical Design

GlobeTrotter is built on a modern MERN stack with optimized build tooling (Vite) and secure protocols.



TECHNICAL IMPLEMENTATION HIGHLIGHTS

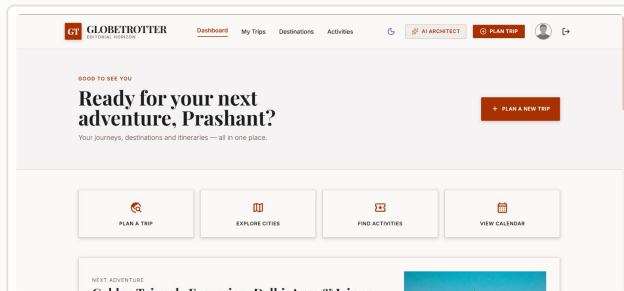
- **Backend:** Node/Express RESTful API handling complex aggregations and sequencing.
- **Security:** JWT for stateless sessions and bcrypt for industry-standard password hashing.
- **Database:** MongoDB used for flexible document-oriented storage of nested trip data.

CONCEPTUAL MONGODB DATA MODEL

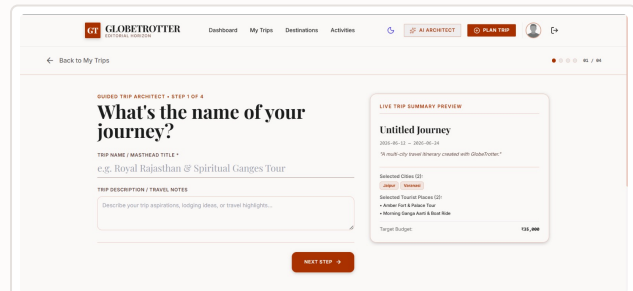
Collection	Purpose
Users	Profiles, hashed credentials, and Google OAuth metadata.
Trips	Root document with name, budget, and ownership details.
Stops	Referenced entities defining specific cities and visit dates.

Implemented Product & UI

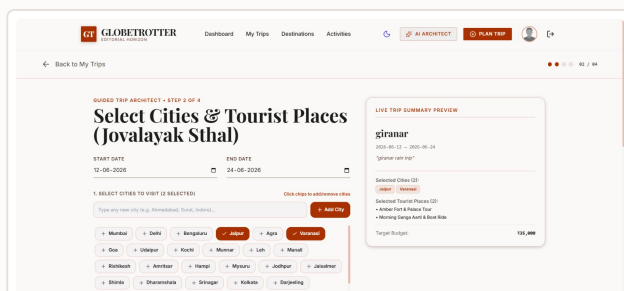
The following screenshots represent the functional GlobeTrotter platform as demonstrated in the final product submission.



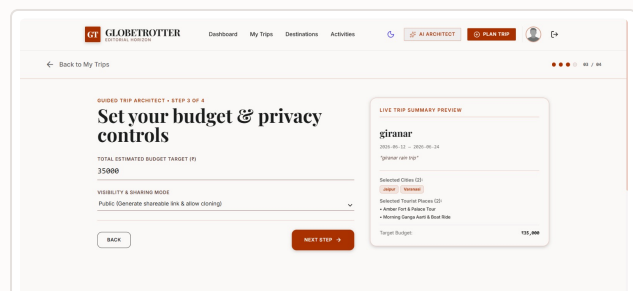
Screenshot 1: Dashboard
Centralized overview for accessing trips, destinations, and activities.



Screenshot 2: Guided Trip Architect
Step-by-step interface for defining journey narrative and trip metadata.



Screenshot 3: City & Tourist Places Selection
Users visually select destinations and local experiences.



Screenshot 4: Budget & Privacy Controls
Interface for defining targets and shareable link visibility.

USER EXPERIENCE ANALYSIS

- **Real-time Preview:** A persistent "Live Trip Summary" updates instantly as cities/activities are added.
- **Editorial Design:** High-quality typography and a burnt-orange brand palette create a premium feel.
- **Guided Navigation:** The 4-step wizard ensures data integrity throughout the planning flow.

Implementation & Technical Impact

KEY TECHNICAL STACK

Technology	Role
React.js	Frontend UI with reusable components.
Node.js	Scalable backend runtime environment.
MongoDB	Flexible document storage for varied itineraries.
JWT	Secure token-based authentication.
Google OAuth	Social login for seamless user onboarding.

PROJECT IMPACT

GlobeTrotter significantly reduces the time required to build a multi-stop itinerary by consolidating discovery and organization. The platform provides financial clarity through target budgeting and encourages social sharing via its unique visibility controls.

SECURITY IMPLEMENTATION

- **Data Protection:** Passwords hashed via bcrypt; JWT stored securely for session management.
- **Authorization:** Protected routes ensure private itineraries remain inaccessible to third parties.

Future Scope & Conclusion

FUTURE ROADMAP

- **AI Integration:** ML-driven activity suggestions based on user behavior and history.
- **Collaboration:** Real-time multi-user editing for family or group trip planning.
- **Map Visualization:** Dynamic route tracing between selected destination stops.
- **Offline Support:** PWA functionality for accessing plans without an internet connection.

CONCLUSION

GlobeTrotter successfully bridges the gap between travel inspiration and organized execution. It provides a unified, secure, and visually stunning platform that simplifies the travel planning lifecycle for the modern explorer.

FINAL SUBMISSION TEAM

Team Name: Cockroach Coders

College: Darshan University

Project Lead: Pritesh Pandya

Team Members: Vala Prashant, Aditya Chauhan, Pandya Karan

GitHub: <https://github.com/prtspndy/GlobeTrotter.git>

DREAM • DESIGN • ORGANIZE • EXPERIENCE